# Basics of Finite Element Simulations for Dynamic Loads & ML-ready Design

## Online Preparation

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## How to use this deck

- This presentation is designed for asynchronous micro-lectures.
- Each section is a self-contained "mini-deck".
- The focus is basic concepts of the Finite Element Method (FEM), immediately usable in real workflows (especially dynamic problems).
- We avoid long derivations and keep the minimum math that clarifies decisions.

**Goal:** being able to set up, check, and interpret FE simulations, and to export clean data for machine learning (ML) models.

# Roadmap

1. **Module A1: The Finite Element Method**
2. **Module A2: Elements & meshing**
3. **Module A3: Nonlinear FEM**
4. **Module A4: Dynamics**
5. **Module A5: Verification, validation, and reporting**
6. **Module A6: FEM $\rightarrow$ dataset for ML**

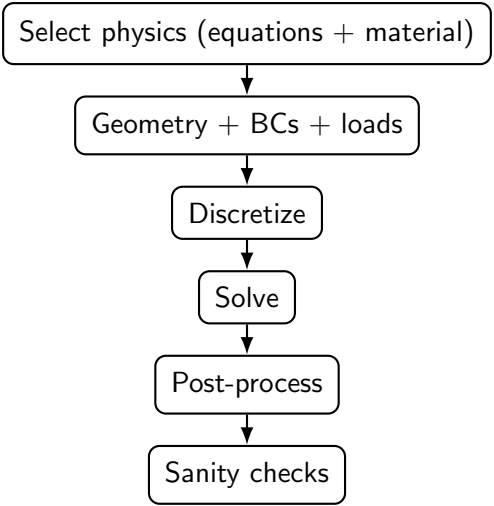
# A1.1 — The FEM in one sentence

**The finite element method (FEM)**

Approximates a continuous field (e.g., displacement) with a finite set of degrees of freedom (DOFs), turning a PDE into a sparse algebraic system.

- Reality: continuum mechanics, fields, PDEs.
- FEM: choose a finite basis $\Rightarrow$ unknown DOFs $\Rightarrow$ solve the discretized linear/nonlinear problem.
- Outputs: displacements $\rightarrow$ strains $\rightarrow$ stresses $\rightarrow$ energies, reactions, failure indicators.

# A1.2 — The pipeline



# A1.3 — A reference structural problem

**Linear elasticity (small strains, small displacements)**

$$\nabla \cdot \boldsymbol{\sigma} + \boldsymbol{f} = 0, \qquad \boldsymbol{\sigma} = \boldsymbol{E}\,\varepsilon, \qquad \varepsilon = \tfrac{1}{2}(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^\top)$$

- The FEM turns this into a sparse system $\boldsymbol{K}\boldsymbol{u} = \boldsymbol{f}$ by discretizing the continuum into discrete elements.
- Inputs: geometry, material ($\boldsymbol{E}$), BCs, loads ($\boldsymbol{f}$).
- Outputs: displacement field $\boldsymbol{u}(\boldsymbol{x})$, stress $\boldsymbol{\sigma}(\boldsymbol{x})$, strain $\varepsilon(\boldsymbol{x})$.

# A1.4 — Degrees of freedom (DOFs)

- Solid elements: DOFs are typically nodal displacements $(u_x, u_y, u_z)$.
- Shell/beam elements: DOFs also include rotations, drilling DOFs, etc.

The FE solution is a vector:

$$\boldsymbol{u} = \begin{bmatrix} \boldsymbol{u_1} & \boldsymbol{u_2} & \cdots & \boldsymbol{u_{n_{\text{dof}}}} \end{bmatrix}^\top$$

**Interpretation**

FEM is not solving everywhere. It solves at DOFs, then interpolates the field inside elements.

# A1.5 — Field approximation: shape functions (1D problem)

Within an element:

$$u(x) = \sum_{i=1}^{n_e} N_i(x)\, u_i$$

- $N_i(x)$: shape functions (local basis), usually linear or quadratic functions.
- $u_i$: nodal DOFs.
- $n_e$: number of nodes of the element.

Strains come from derivatives:

$$\varepsilon(x) = \boldsymbol{B}(x)\, \boldsymbol{u}_e$$

where $\boldsymbol{B}$ contains derivatives of shape functions and $\boldsymbol{u}_e$ is the vector of nodal displacements.

# A1.6 — Element stiffness and forces

For a linear elastic solid element:

$$\boldsymbol{K}_e = \int_{\Omega_e} \boldsymbol{B}^\top \boldsymbol{E}\, \boldsymbol{B}\, \mathrm{d}\Omega \qquad \boldsymbol{f}_e = \int_{\Omega_e} \boldsymbol{N}^\top \boldsymbol{b}\, \mathrm{d}\Omega + \int_{\Gamma_e} \boldsymbol{N}^\top \boldsymbol{t}\, \mathrm{d}\Gamma$$

- $\Omega_e$: Volume of the element.
- $\Gamma_e$: surface of the element.
- $\boldsymbol{b}$: body forces.
- $\boldsymbol{t}$: surface forces.

Similar equations describe mass ($\boldsymbol{M}$) and damping ($\boldsymbol{D}$) matrices of elements.
Integrals computed numerically (Gauss quadrature).

# A1.7 — Assembly: local information $\rightarrow$ global system

- Each element has local DOFs; assembly maps them into global indices.
- The global matrix is sparse because elements connect only nearby nodes.

**Outcome for a static problem**

$$\boldsymbol{K}\boldsymbol{u} = \boldsymbol{f}$$

with $\boldsymbol{K}$ sparse.

# A1.8 — Boundary conditions

Boundary conditions must be introduced in order to make the problem well-posed and solvable.
Two main options are available: prescribed displacements and loads.

**Prescribed displacement**

$$u = \bar{u} \quad \text{on } \Gamma_u$$

- Removes rigid-body motion

**Distributed loads**

$$\sigma \cdot n = t \quad \text{on } \Gamma_t$$

- Enters via surface integrals

Other options are available, among which initial velocity and concentrated forces are mostly used.

# A1.9 — Results and post-processing

- Displacements are primary unknowns; everything else is derived.
- Stresses typically computed at Gauss points then extrapolated / averaged.
- Derived quantities can be sensitive to mesh and singularities.

**Prefer robust quantities for decisions**

- global reactions, compliance/strain energy, displacement
- energy balance (especially in dynamics)
- averaged or integrated stress measures (not peak at a corner)

# A1.12 — Common pitfalls (and why they happen)

- **Under-constraint:** rigid-body modes $\rightarrow$ singular $\boldsymbol{K}$ or drifting solution.
- **Over-constraint:** artificial stiffness, unrealistic stress patterns.
- **Stress singularities:** point loads, sharp re-entrant corners, perfectly rigid constraints.
- **Misleading "max stress":** mesh-dependent peaks near singularities.

**Practical fix**

Use realistic load distributions, fillets, contact areas; evaluate stresses away from singular spots.

# A2.1 — Element families

### 1D / 2D / 3D

- 1D: slender members, fast
- 2D: thin structures (thickness $\ll$ in-plane dimension)
- 3D: general 3D structures, heavier cost

### Order

- Linear: cheaper, can be too stiff in bending
- Quadratic: better curvature, more DOFs

**Decision lens:** geometry + dominant deformation mode + required outputs.

# A2.2 — Gauss integration

- Integrals are evaluated at Gauss points.
- Full integration: robust, can lock (too stiff) in certain cases.
- Reduced integration: cheaper, can introduce zero-energy deformation modes (hourglassing) if not controlled.

**Practical message**

If an element formulation exists in standard and reduced versions, know the trade: reduced can help locking, but needs stabilization/controls.

## A2.3 — Element/Mesh density

Mesh density should be increased, that is, element size should be decreased, where physics changes:

- High gradients: holes, fillets, contact zones, load introduction regions.
- Wave phenomena (dynamic): need enough elements per wavelength.

If you only refine globally, you pay huge cost for little accuracy.

**Engineering heuristic**

Refine where geometry curvature, loads/BC changes, or contacts create steep gradients.

## A2.4 — Mesh convergence

Mesh convergence studies are necessary to verify that the numerical solution becomes insensitive to further mesh refinement.

- Global quantities (stiffness/compliance, strain energy, reaction forces) converge earlier.
- Local peak stresses near singularities may never converge meaningfully.
- For dynamics, also check time-step sensitivity (not just mesh).

**Minimum convergence workflow**

1. pick 1–3 key KPIs (e.g., max deflection, absorbed energy),
2. apply Richardson's extrapolation,
3. identify the element size that balances accuracy and computational time.

# A2.5 — Element distortion and mesh quality

- Highly skewed/distorted elements degrade accuracy and can break convergence.
- Bad quality is especially dangerous in:
  - contact problems,
  - explicit dynamics (tiny stable time step),
  - large deformation.

**Practical symptoms**

Nonphysical oscillations, negative Jacobians, solver warnings, energy drift, "checkerboard" stress patterns.

## A2.6 — Common forms of locking

Element selection must account for the possibility that certain element formulations may suffer from shear or volumetric locking under specific geometric or material conditions.

### Shear locking (bending with low-order elements)

Element becomes too stiff in bending $\Rightarrow$ underestimates deflections.

### Volumetric locking (nearly incompressible response)

Artificial constraint on volume change $\Rightarrow$ overly stiff response.

**Typical fixes:**

- use higher-order elements or mixed formulations,
- consider reduced integration with stabilization,
- verify on a benchmark (simple bending / incompressible test).

## A2.7 — Hourglass modes

Reduced integration elements may exhibit zero-energy deformation modes (hourglass modes) if not properly stabilized.

### Origin (under-integrated low-order elements)

Insufficient integration $\Rightarrow$ spurious zero-energy modes not detected by the stiffness matrix.

### Effect on the solution

Non-physical deformation patterns $\Rightarrow$ mesh-dependent results or unstable response.

**Typical fixes:**

- use appropriate hourglass control or stabilization schemes,
- prefer fully integrated or higher-order elements when suitable,
- monitor artificial energy contributions in the solution.

# A3.1 — What makes a FEM problem nonlinear?

- **Material nonlinearity:** plasticity, viscous behavior, damage.
- **Geometric nonlinearity:** large deformation/strain.
- **Contact nonlinearity:** boundary conditions change as bodies touch/separate.

**Why you care**

Dynamic loads often involve plasticity + contact + large deformation.

# A3.2 — Residual form: equilibrium for static or implicit dynamic

Nonlinear equilibrium at a time step:

$$\boldsymbol{R}(\boldsymbol{u}) = \boldsymbol{f}_{\mathrm{ext}} - \boldsymbol{f}_{\mathrm{int}}(\boldsymbol{u}) = \boldsymbol{0}$$

Newton-type iterations:

$$\boldsymbol{K}_t(\boldsymbol{u}^{(k)})\Delta\boldsymbol{u}^{(k)} = \boldsymbol{R}(\boldsymbol{u}^{(k)}), \quad \boldsymbol{u}^{(k+1)} = \boldsymbol{u}^{(k)} + \Delta\boldsymbol{u}^{(k)}$$

**Interpretation**

Convergence means the model found a configuration where internal forces balance external forces (within tolerances).

## A3.3 — Convergence criteria

Solution convergence is typically evaluated according to one of the following criteria:

1. Residual norm $\boldsymbol{R}(\boldsymbol{u})$ decreases (force balance).
2. Increment norm $\Delta\boldsymbol{u}$ decreases (solution stabilizes).
3. Energy norm $\boldsymbol{R}(\boldsymbol{u}) \cdot \Delta\boldsymbol{u}$ is consistent (work balance).

**Common causes of non-convergence**

- contact setup too stiff/abrupt,
- time/load step too large,
- element distortion,
- material parameters inconsistent with units.

# A3.4 — Plasticity: the minimum concepts you need

- Elastic regime: $\sigma = \boldsymbol{D}\varepsilon$.
- Yield condition: $f(\sigma, \kappa) \leq 0$ (e.g., von Mises).
- Flow rule: plastic strain evolves when $f = 0$.
- Hardening: yield surface expands/translates (material strengthens/softens).

**Outputs that matter in dynamics**

Plastic strain, plastic work/dissipation, absorbed energy, permanent deformation.

# A3.5 — Contact

- Contact switches constraints on/off (non-smooth behavior).
- Typical enforcement:
  - **Penalty:** allows small penetration; stable and simple.
  - **Lagrange multipliers:** enforces no-penetration; adds DOFs; can be harder.
- Friction adds nonlinearity (stick/slip transitions).

**Practical advice**

Start simple (frictionless), validate, then add friction and complexity.

# A4.1 — Dynamic FEM: equation of motion

Semi-discrete structural dynamics:

$$\boldsymbol{M}\ddot{\boldsymbol{u}}(t) + \boldsymbol{C}\dot{\boldsymbol{u}}(t) + \boldsymbol{K}\boldsymbol{u}(t) = \boldsymbol{f}(t)$$

- $\boldsymbol{M}$: mass matrix.
- $\boldsymbol{C}$: damping.
- $\boldsymbol{K}$: stiffness.
- $\boldsymbol{f}(t)$: time-dependent loads.

# A4.2 — Implicit vs explicit integration

**Implicit (e.g., Newmark)**

- solves a system each step
- can use larger $\Delta t$
- good for low-frequency, quasi-static dynamics

**Explicit (central difference)**

- no global solve per step (fast per step)
- stability-limited $\Delta t$
- ideal for short, violent events (impact/blast)

**Key:** explicit cost is dominated by number of steps $\sim T/\Delta t$.

# A4.3 — Explicit stability

For explicit dynamics, a stability limit is linked to wave propagation:

$$\Delta t_{\text{crit}} \sim \frac{L_{\min}}{c} \qquad c \approx \sqrt{\frac{E}{\rho}}$$

- $L_{\min}$: smallest characteristic element length.
- $c$: wave speed in the material.
- Refining mesh in a tiny region can reduce $\Delta t$ for the whole model.

**Practical implication**

Mesh is not just accuracy; in explicit dynamics it is also runtime.

# A4.4 — Damping

Real materials have damping, but it is difficult to model precisely. Rayleigh (proportional) damping is often used in FE simulations:

$$\boldsymbol{C} = \alpha \boldsymbol{M} + \beta \boldsymbol{K}$$

Numerical damping may also arise from algorithms or stabilization.

**Warning**

Too much damping can solve numerical noise by deleting real physics.

# A4.5 — Energy

Energy must be tracked in explicit simulations to ensure there are no errors. Typical energy variables to track:

- Kinetic energy $E_k$
- Internal energy (strain + plastic work) $E_{\text{int}}$
- Contact/friction energy
- Hourglass / artificial energies (if applicable)

**What you want to see**

Energy evolves smoothly; no unexplained blow-up; energy balance consistent with applied work/impulse; artificial energy less than 10% of the internal energy.

# A4.6 — Outputs for dynamic problems

Some key outputs you want to check:

- Time histories at key points: $u(t)$, $\dot{u}(t)$, $\ddot{u}(t)$.
- Integrated variables: absorbed energy, transmitted impulse.
- Field snapshots at meaningful times.

**Avoid**

Relying on a single max stress value without context; it is often a mesh artifact.

# A5.1 — Verification vs validation (V&V)

Simulations must be verified and validated to ensure they can replicate real physics.

**Verification**

Did we solve the equations right? (discretization, solver, convergence, implementation)

**Validation**

Did we pick the right equations? (model assumptions vs reality, experiments, calibration)

- Focus heavily on **verification** and on **defensible modeling**.
- Validation is ideal but often limited by experimental data availability.

# A5.2 — Checks

At least six checks should be made before trusting simulations:

1. Units and magnitudes (materials, density, loads)
2. Constraints (rigid-body modes? over-constraint?)
3. Equilibrium (reactions vs loads / impulse)
4. Mesh sensitivity
5. Time-step sensitivity
6. Energy balance

## A5.3 — Minimum reporting standard

When documenting numerical simulation, one must address:

- Geometry + coordinate system + simplifications (symmetry, submodeling).
- Element type, integration, mesh sizes in critical zones.
- Material model + parameters + units.
- Loads/BCs (including time histories) and contact definitions.
- Solver settings (implicit/explicit, tolerances, step sizes).
- Verification evidence (mesh/time sensitivity, energy plots, equilibrium).
- Validation evidence (comparison with exeperimental data)

# A6.1 — Datapoint in simulation-driven design

**Datapoint**

A single, discrete unit of information or observation within a larger dataset. Usually, it comes from an unknown map **input** $\rightarrow$ **output**.

- Inputs: geometry parameters, material parameters, boundary conditions, load parameters.
- Outputs: scalar performance indicators (for example, max deflection), curves ($u(t)$), fields ($\sigma(x)$).

## A6.2 — Datapoint choice in dynamic FE simulations

Not all variables can be used to create ML datasets. For ML, you want outputs that are stable, repeatable, and physically meaningful.

- **Good labels (robust):**
  - absorbed internal energy / plastic work,
  - impulse transmitted to supports,
  - max deflection (with location defined),
  - permanent deformation.

- **Risky labels (often mesh/singularity sensitive):**
  - peak von Mises at a sharp corner,
  - stress at a point load application node,
  - unconstrained max over the entire mesh.

# A6.3 — Datapoint choice in dynamic FE simulations - cont.

Time histories are usually condensed into single-valued indicators. For example, in case of pressure load $p(t)$:

- Peak overpressure,
- Duration,
- Impulse $I = \int p(t)\mathrm{d}t$.

For structural response, energy and displacement are usually considered:

- Maximum displacement $u_{\mathrm{max}}$,
- time-to-peak,
- rebound ratio,
- absorbed energy,
- residual deflection.

# A6.4 — Data processing to improve learning

- Normalize/scaling: track units and apply consistent scaling.
- Train/Validation/test splits should respect **families** (e.g., geometry groups) to test generalization.
- Log everything: mesh size, solver settings, time step, material version.

**Engineering truth**

A surrogate learns your pipeline assumptions as much as it learns mechanics.

# A6.5 — Suggested metadata scheme

| Field | Example |
|---|---|
| geometry params | lattice cell size, thickness, fillet radius |
| material params | E, $\nu$, $\rho$, yield, hardening, rate params |
| BCs | fixed edge, symmetry planes |
| load params | peak overpressure $p_{\max}$, duration $\tau$, impulse $I$ |
| solver params | explicit/implicit, $\Delta t$, element type, mesh stats |
| outputs | $u_{\max}$, absorbed energy, transmitted impulse |
| checks | equilibrium error, energy drift |